

# GSHTrans: mathematics and numerical methods

A reference for what the library does, and why

## Abstract

This is a record rather than a plan. Part I collects the definitions, conventions and relations the library rests on, in the form it actually uses them, with a translation table for the three sources whose notations differ. Part II explains the numerical methods: what the algorithms are, why they were chosen, and the handful of places where the reasoning is subtle enough to be worth writing down. Neither part is a derivation. The companion note `canonical-components.tex` carries the mathematical arguments; `core-plan.md` and `field-algebra-plan.md` carry the design decisions and the measurements behind them.

Statements here are either verified by a test in the library or read directly from a cited source. The two places where a convention remains genuinely open are marked as such.

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## Part I

# Mathematics

## 1 Conventions

### 1.1 The canonical basis

At each point of the sphere, with  $\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}}$  the usual orthonormal spherical polar frame,

$$\mathbf{e}_{\pm} = \mp \frac{1}{\sqrt{2}} \left( \hat{\boldsymbol{\theta}} \pm i \hat{\boldsymbol{\phi}} \right), \quad \mathbf{e}_0 = \hat{\mathbf{r}}, \quad \alpha \in \{-1, 0, +1\}. \quad (1)$$

Two consequences are used everywhere:

$$\overline{\mathbf{e}_{\alpha}} = (-1)^{\alpha} \mathbf{e}_{-\alpha}, \quad g_{\alpha\beta} = \mathbf{e}_{\alpha} \cdot \mathbf{e}_{\beta} = (-1)^{\alpha} \delta_{\alpha+\beta, 0}. \quad (2)$$

So  $g_{+-} = g_{-+} = -1$ ,  $g_{00} = 1$ , and  $g$  is its own inverse. The library works with contravariant components throughout and applies  $g$  explicitly wherever a contraction is taken; index position is not tracked.

Under a rotation of the tangent frame through  $\psi$  about  $\hat{\mathbf{r}}$ ,  $\mathbf{e}_{\pm} \mapsto e^{\mp i\psi} \mathbf{e}_{\pm}$ .

## 1.2 What the library stores, exactly

The single most useful statement for reading the code:

Wigner’s stored value at upper index  $N$ , degree  $l$  and order  $m$  is

$$\left(\frac{2l+1}{4\pi}\right)^{1/2} d_{Nm}^l(\theta) = X_{lm}^N(\theta), \quad (3)$$

which is exactly the scalar harmonic  $X_{lm}^N$  of Dahlen & Tromp (1998), their (C.117), built on their generalized Legendre function  $P_{lm}^N = d_{Nm}^l$  — *upper index first*.

This is verified rather than assumed. `tests/CheckWignerConvention.h` compares all nine  $l = 1$  values against D&T (C.115); the check discriminates, because the four entries with  $N - m$  odd distinguish  $d_{Nm}^l$  from  $d_{mN}^l$ , which differ by  $(-1)^{N-m}$ . Two further facts corroborate it: the recursion seed evaluates to D&T (C.125), and `tests/CheckLegendre.h` pins the  $N = 0$  row against `std::sph_legendre`, fixing the orthonormal scaling and the Condon–Shortley phase.

## 1.3 Translation table

The three standard sources use different notations, and two of them differ from this library in normalisation. Everything in this table is read directly from the cited equations.

|                          | harmonic                         | colatitudinal part                                         | normalised? |
|--------------------------|----------------------------------|------------------------------------------------------------|-------------|
| GSHTrans                 | $Y_{lm}^N = X_{lm}^N e^{im\phi}$ | $X_{lm}^N = \left(\frac{2l+1}{4\pi}\right)^{1/2} d_{Nm}^l$ | yes         |
| Dahlen & Tromp (C.117)   | $Y_{lm}^N = X_{lm}^N e^{im\phi}$ | $X_{lm}^N = \left(\frac{2l+1}{4\pi}\right)^{1/2} P_{lm}^N$ | yes         |
| Woodhouse & Deuss [41]   | $Y_l^{Nm} = P_l^{Nm} e^{im\phi}$ | $P_l^{Nm} = d_{Nm}^{(l)}$                                  | <b>no</b>   |
| Phinney & Burridge (1.5) | $X^{Nm} = P_l^{Nm} e^{im\phi}$   | $P_l^{Nm}$ , their Appendix A                              | <b>no</b>   |

Three things to take from it.

**The index order is the same in all four.** D&T write  $P_{lm}^N$ , Woodhouse and P&B write  $P_l^{Nm}$ , and this library writes  $d_{Nm}^l$ ; all mean the same function with the upper index first. D&T state that they follow P&B in the definition.

**The normalisation is not.** D&T and this library carry the  $[(2l+1)/4\pi]^{1/2}$ ; Woodhouse and P&B do not. A formula taken from Woodhouse or P&B therefore needs that factor supplied, and a formula taken from D&T does not.

**The transposed function is a different thing.** D&T (C.118) gives  $P_{lN}^m = (-1)^{m+N} P_{lm}^N$ , so getting the index order wrong costs a sign whenever  $m - N$  is odd, and nothing when it is even — which is exactly the kind of error that survives half a test suite.

## 1.4 What remains a convention

Two statements in this document cannot be settled from inside the library.

1. The sign  $\mp$  in (1). Some authors use  $\mathbf{e}_{\pm} = \pm(\hat{\boldsymbol{\theta}} \pm i\hat{\boldsymbol{\phi}})/\sqrt{2}$ , which flips every odd-rank reality phase. Nothing the library currently computes observes it.
2. The overall signs of  $\tilde{\mathfrak{d}}$  and  $\overline{\tilde{\mathfrak{d}}}$  (§6). The magnitudes are not in doubt, and two identities pin everything else, but both survive flipping the pair.

## 2 Canonical components

A rank- $p$  tensor field has  $3^p$  scalar components, each labelled by a multi-index  $(\alpha_1 \dots \alpha_p) \in \{-1, 0, 1\}^p$ . Under the frame rotation, the component  $T^{\alpha_1 \dots \alpha_p}$  acquires the phase  $e^{-iN\psi}$  with

$$N = \sum_{i=1}^p \alpha_i, \quad (4)$$

and it is this  $N$  — not the multi-index — that is the upper index of the component’s harmonic expansion.

**The multi-index is not the upper index.** For a vector they coincide,  $N = \alpha$ , which is the coincidence that misleads. For  $p \geq 2$  distinct components share an upper index: a rank-2 tensor has three at  $N = 0$ , namely  $(-+)$ ,  $(00)$  and  $(+-)$ . A collection of components labelled only by  $N$  therefore does not determine a tensor. The number carrying a given  $N$  is the trinomial coefficient; for  $p = 2$  these are 1, 2, 3, 2, 1 across  $N = -2 \dots 2$ .

**Products and contractions.** Concatenating multi-indices adds upper indices by (4), which is the statement “upper indices add under pointwise multiplication”. Contracting slots  $j$  and  $k$  against the metric gives

$$(\text{tr}_{jk} T)^{\dots} = \sum_{\alpha} (-1)^{\alpha} T^{\dots \alpha \dots -\alpha \dots} = -T^{\dots - \dots + \dots} + T^{\dots 0 \dots 0 \dots} - T^{\dots + \dots - \dots}, \quad (5)$$

the contracted pair contributing  $\alpha + (-\alpha) = 0$ , so the result carries the upper index of the surviving slots.

## 3 Generalized spherical harmonics

A field  $f$  of upper index  $N$  expands as

$$f(\theta, \phi) = \sum_{l \geq |N|} \sum_{m=-l}^l f_{lm}^N Y_{lm}^N, \quad f_{lm}^N = \int_{S^2} \overline{Y_{lm}^N} f \, d\Omega, \quad (6)$$

with

$$Y_{lm}^N(\theta, \phi) = \left( \frac{2l+1}{4\pi} \right)^{1/2} d_{Nm}^l(\theta) e^{im\phi}. \quad (7)$$

These are orthonormal,  $\int \overline{Y_{lm}^N} Y_{l'm'}^N \, d\Omega = \delta_{ll'} \delta_{mm'}$ . The sum starts at  $l = |N|$  because  $d_{Nm}^l$  vanishes identically below it: there is no harmonic of that upper index at lower degree, and correspondingly no coefficient to store. This is the single fact most likely to look like a bug on first contact — see §9.3.

**The normalisation is pinned by the code.** Taking  $f$  constant, (6) gives  $f_{00}^0 = 2\sqrt{\pi}f$ , and `GaussLegendreGrid` implements exactly that on its one-point grid.

**Conjugation.** Using  $d_{-m, -N}^l = (-1)^{m-N} d_{mN}^l$  and the reality of  $d$ , equation (7) gives D&T (C.109),

$$\overline{Y_{lm}^N} = (-1)^{m+N} Y_{l, -m}^{-N}, \quad (8)$$

from which follow the two spectral reality relations of §5.

## 4 The generalized Legendre functions

$P_{lm}^N(\mu) = d_{Nm}^l(\theta)$ ,  $\mu = \cos \theta$ , is the regular solution of the generalized Legendre equation, normalised as D&T (C.112) so that the addition theorem below holds. It does *not* reduce to the associated Legendre function at  $N = 0$ ; D&T (C.113) relates them.

### 4.1 Symmetries

D&T (C.118), in full, since the library uses two of them and could use two more:

$$\begin{aligned} P_{l,-m}^N(\mu) &= (-1)^{l+N} P_{lm}^N(-\mu), & P_{lm}^{-N}(\mu) &= (-1)^{l+m} P_{lm}^N(-\mu), \\ P_{l,-m}^{-N}(\mu) &= (-1)^{m+N} P_{lm}^N(\mu), & P_{lN}^m(\mu) &= (-1)^{m+N} P_{lm}^N(\mu), \\ P_{l,-N}^{-m}(\mu) &= P_{lm}^N(\mu). \end{aligned} \tag{9}$$

The third is the  $(m, N) \mapsto (-m, -N)$  involution; the second is the  $\theta \mapsto \pi - \theta$  reflection, which maps the Gauss–Legendre node set to itself exactly, since those nodes are symmetric about  $\pi/2$ . Composed, the two generate a group of order four acting on  $(N, m, \theta)$ , so the table of stored values could in principle be reduced fourfold. Neither reduction is implemented; §15 says why. The fourth relation is the transposition that the convention check discriminates against.

### 4.2 Boundary values

The values at the extreme orders have closed forms, D&T (C.123) and (C.125), and these are precisely the library’s `WignerMinOrder` and `WignerMaxOrder`:

$$P_{l,-l}^N = \left[ \frac{(2l)!}{(l+N)!(l-N)!} \right]^{1/2} \left( \sin \frac{\theta}{2} \right)^{l+N} \left( \cos \frac{\theta}{2} \right)^{l-N}, \tag{10}$$

$$P_{ll}^N = (-1)^{l+N} P_{l,-l}^N|_{N \rightarrow -N}. \tag{11}$$

They are the seeds of D&T’s recursion in  $m$ , and the boundary terms of the library’s recursion in  $l$  (§10).

### 4.3 The addition theorem

D&T (C.127),

$$\sum_{m=-l}^l P_{lm}^N(\mu) P_{lm}^{N'}(\mu) = \delta_{NN'}. \tag{12}$$

This is the strongest available check on a computed table, and `tests/CheckAdditionTheorem.h` is it. Because the library stores  $X_{lm}^N$  rather than  $P_{lm}^N$ , the right-hand side there is  $\delta_{NN'}(2l+1)/4\pi$ .

## 5 Reality

For a real tensor field, applying (2) to each factor gives

$$T^{-\alpha_1 \dots -\alpha_p} = (-1)^N \overline{T^{\alpha_1 \dots \alpha_p}}. \tag{13}$$

For a vector this reads  $u^0 = \overline{u^0}$  and  $u^- = -\overline{u^+}$ : the radial component is a real field and the two transverse components are not independent.

**Which components must be stored.** (13) pairs each multi-index with its negation, and one representative of each orbit suffices. The only fixed point is the all-zero index, which is therefore a real-valued field; every other orbit has size two. With a permutation symmetry present the orbits are those of the group generated by the permutations *together with* the negation, and a component whose orbit contains its own negation is pinned to a single real number — real, or purely imaginary when a permutation sign intervenes. Such self-paired components always sit at  $N = 0$ : permutation preserves the slot sum and negation reverses it, so a component fixed by the combination satisfies  $N = -N$ .

The resulting counts, computed by the library rather than tabulated by hand, are the independent real degrees of freedom in every case:

|                       | complex components | pinned | reals per point |
|-----------------------|--------------------|--------|-----------------|
| rank 1                | 1                  | 1      | 3               |
| rank 2                | 4                  | 1      | 9               |
| rank 2, symmetric     | 2                  | 2      | 6               |
| rank 2, antisymmetric | 1                  | 1      | 3               |
| rank 3, symmetric     | 4                  | 2      | 10              |
| rank 4                | 40                 | 1      | 81              |
| rank 4, elastic       | 8                  | 5      | 21              |

The last row is worth noting: nothing in the machinery knows about elasticity, and  $c_{ijkl} = c_{jikl} = c_{ijlk} = c_{klij}$  comes out at the familiar 21.

**In the spectral domain** the same relation takes two distinct forms, and it is worth keeping them apart. For *any* field  $f$  of upper index  $N$ , (8) gives the coefficients of  $\bar{f}$ , which has upper index  $-N$ :

$$(\bar{f})_{l,-m}^{-N} = (-1)^{m-N} \overline{f_{lm}^N}. \quad (14)$$

For a negation-paired component of a *real tensor*, (13) supplies a further  $(-1)^N$  and the two combine to

$$T_{l,-m}^{-N} = (-1)^m \overline{T_{lm}^N}, \quad (15)$$

which at  $N = 0$  is the familiar condition on a real scalar field, and is what the reduced  $m \geq 0$  coefficient storage implements.

## 6 Raising and lowering

$\partial$  and  $\bar{\partial}$  change the upper index by one. On a basis function,

$$\partial Y_{lm}^N = -\sqrt{(l-N)(l+N+1)} Y_{lm}^{N+1}, \quad \bar{\partial} Y_{lm}^N = +\sqrt{(l+N)(l-N+1)} Y_{lm}^{N-1}, \quad (16)$$

so in the spectral domain each is a multiplication by an  $l$ -dependent factor, with no coupling between different  $(l, m)$ . They are the operations that connect different upper indices, and they are diagonal in  $(l, m)$  rather than local in  $(\theta, \phi)$  — which is why the library has two representations rather than one, and why an expression such as “the gradient of a product” is necessarily evaluated in both.

The surface gradient of a scalar has canonical components proportional to  $\partial f$  and  $\bar{\partial} f$ , at upper index  $+1$  and  $-1$ , with vanishing radial component.

**Two identities, and what they do not pin.** Composing the factors,

$$\bar{\partial}\partial|_{N=0} = -l(l+1), \quad [\bar{\partial}, \partial] = -2N. \quad (17)$$

The first says  $\bar{\partial}\partial$  is the surface Laplacian on a scalar; it pins the relative sign of the two operators and both magnitudes. The second is structural. Both are tested. **Both survive flipping the sign of the pair**, so the overall sign in (16) remains a convention to be fixed against Phinney & Burridge, and is implemented as stated above.

**The degree ranges look after themselves.** Raising a field at  $N \geq 0$  produces one starting at  $l = N + 1$ , and the factor at  $l = N$  is  $\sqrt{0} = 0$ : the coefficient with nowhere to go was going to vanish. Lowering produces a field starting at a lower degree, whose extra coefficient has no source and is zero, which is what having no content there means.

## 7 The contravariant derivative, and how it differs

Phinney & Burridge, and Dahlen & Tromp following them, work not with  $\bar{\partial}$  but with a *contravariant derivative*  $\partial^\sigma$ ,  $\sigma \in \{-, 0, +\}$ , which takes the expansion coefficients of a rank- $q$  tensor to those of the rank- $(q+1)$  tensor  $\nabla \mathbf{T}$ . It is closely related to  $\bar{\partial}$  and it is not the same operator, and the difference matters for anyone using this library on tensors.

### 7.1 The definition

With D&T (C.146),

$$\Omega_l^{\pm N} = \sqrt{\frac{1}{2}(l \pm N)(l \mp N + 1)}, \quad (18)$$

their (C.151)–(C.153) read

$$\partial^- T_{lm}^{\alpha_1 \dots \alpha_q} = r^{-1} \left[ \Omega_l^{+N} T_{lm}^{\alpha_1 \alpha_2 \dots \alpha_q} - T_{lm}^{(\alpha_1 - 1) \alpha_2 \dots \alpha_q} - \dots - T_{lm}^{\alpha_1 \alpha_2 \dots (\alpha_q - 1)} \right], \quad (19)$$

$$\partial^0 T_{lm}^{\alpha_1 \dots \alpha_q} = \frac{dT_{lm}^{\alpha_1 \dots \alpha_q}}{dr}, \quad (20)$$

$$\partial^+ T_{lm}^{\alpha_1 \dots \alpha_q} = r^{-1} \left[ \Omega_l^{-N} T_{lm}^{\alpha_1 \alpha_2 \dots \alpha_q} - T_{lm}^{(\alpha_1 + 1) \alpha_2 \dots \alpha_q} - \dots - T_{lm}^{\alpha_1 \alpha_2 \dots (\alpha_q + 1)} \right], \quad (21)$$

with the stipulation that any coefficient whose shifted index leaves  $\{-1, 0, 1\}$  is zero. The components of the gradient are  $(\nabla \mathbf{T})^{\sigma \alpha_1 \dots \alpha_q} = \partial^\sigma T^{\alpha_1 \dots \alpha_q}$ : the operator *prepends a slot*, and the result carries upper index  $\sigma + N$ , which is (4) applied to the lengthened multi-index.

### 7.2 Two differences, one benign and one not

**A factor of  $\sqrt{2}$ , which is only the basis normalisation.** Comparing (18) with (16),

$$|\text{raising factor}| = \sqrt{(l - N)(l + N + 1)} = \sqrt{2} \Omega_l^{-N}, \quad |\text{lowering factor}| = \sqrt{2} \Omega_l^{+N}. \quad (22)$$

The  $\sqrt{2}$  is exactly the  $1/\sqrt{2}$  in  $\mathbf{e}_\pm = \mp(\hat{\boldsymbol{\theta}} \pm i\hat{\boldsymbol{\phi}})/\sqrt{2}$ , which appears explicitly in D&T's surface gradient (C.144). So on the unit sphere, up to that factor and to the sign convention of §6,  $\partial^+$  is  $\bar{\partial}$  and  $\partial^-$  is  $\bar{\partial}$  — *for a scalar*.

**Connection terms, which are not benign.** For  $q \geq 1$  the bracket in (19) and (21) is not a multiplication. It subtracts the components with one slot index shifted, once per slot. Those terms are the connection: the canonical basis vectors themselves vary over the sphere,

$$[\pm \partial_\theta + i(\sin \theta)^{-1} \partial_\phi] \mathbf{e}_\alpha = \alpha \cot \theta \mathbf{e}_\alpha - \sqrt{2} \mathbf{e}_{\alpha \pm 1}, \quad (23)$$

so differentiating a tensor field differentiates its basis too. A scalar has no slots and therefore no connection terms, which is why the two operators coincide there and only there.

### 7.3 The consequence for this library

**Applying  $\partial$  to each component of a tensor does not give the gradient of the tensor.** Use `SurfaceGradient`. It gives the  $\partial$  of each component as a spin-weighted function, which for  $q \geq 1$  is not a component of  $\nabla \mathbf{T}$  and in general is not a component of anything.

This is a trap rather than a limitation: `Raise` and `Lower` are correct for what they claim — a single spin-weighted function — and are correct for the surface gradient of a scalar, which is the case most people reach for first. They are simply not the tensor operator, and nothing in their types says so, because a tensor expansion’s components are ordinary spin expansions.

### 7.4 What is implemented

`SurfaceGradient` is  $\nabla_1$ : (19) and (21) without the  $r^{-1}$ , taking a rank- $q$  tensor expansion to a rank- $(q+1)$  one.

**No radial part.**  $\partial^0$  is  $\partial/\partial r$ , which an angular library cannot supply; but the *surface* gradient has no  $\mathbf{e}_0$  component at all (D&T C.145), so the operator is complete without it and the radial derivative is the user’s to provide. What that means for the result is that its  $\sigma = 0$  block is stored and zero: a rank- $(q+1)$  tensor whose  $\mathbf{e}_0$  slot vanishes is an ordinary tensor, and keeping it as one leaves the result composable with contraction, further gradients and the transform, for a third of the storage.

**The result has no symmetry and keeps the operand’s reality.**  $\nabla_1$  of a symmetric tensor is not symmetric in the new slot against the old ones, and inferring a symmetry from an operator is the problem `Materialise` declined to solve.  $\nabla_1$  of a real tensor is real, so only the reduced set is computed.

**It needed one thing phase 5 had left out.** Every term is a coefficient of some component at fixed  $(l, m)$ , and the shifted components need not be stored. `TensorExpansion::Coefficient` now answers for *any* representable component, resolving the stored case, a permutation relative, a reality relative by (15), a real block’s reduced  $m \geq 0$  storage, a pinning as imaginary, and a vanishing orbit. It is the spectral counterpart of phase 4’s component accessor and closes the asymmetry noted above.

**What the tests pin.** On a scalar the operator is  $\partial$  up to  $\sqrt{2}$ , which checks the  $\Omega$  factors and nothing else. The metric trace of the *second* gradient of a scalar is the surface Laplacian,  $-l(l+1)$  — and that runs through the connection terms, since  $\partial^-$  of the  $+1$  component subtracts the 0 component, so the identity fails outright if the operator is  $\partial$  applied component-wise. The chain rule (24) is checked at rank one, in the spatial domain. And the gradient of a real tensor is checked against the same field widened to a complex one, which exercises reading derived components and writing pinned ones.

**Divergence and curl** are contractions of the gradient (D&T C.6.2). Contraction is pointwise, so a caller evaluates the gradient and contracts with the machinery of §13; there is no separate operator.

**One further property worth knowing.**  $\partial^\sigma$  preserves the chain rule as an operator on fields, D&T (C.154)–(C.155):  $\nabla(\mathbf{TP}) = (\nabla \mathbf{T})\mathbf{P} + \mathbf{T}(\nabla \mathbf{P})$  gives

$$\partial^\sigma (T^{\alpha\dots} P^{\beta\dots}) = (\partial^\sigma T^{\alpha\dots}) P^{\beta\dots} + T^{\alpha\dots} (\partial^\sigma P^{\beta\dots}). \quad (24)$$



It does *not* hold coefficient by coefficient, because a product of fields is not a product of coefficients — which is the same fact that makes “the gradient of a product” an operation in two representations.

## 8 Quadrature

**Colatitude.** Gauss–Legendre with  $n_\theta = l_{\max} + 1$  nodes, which integrates a polynomial of degree  $2n_\theta - 1$  in  $\cos \theta$  exactly. The nodes are the roots of  $P_{n_\theta}$ , mapped by  $\theta = \arccos(-x)$ , and are symmetric about  $\pi/2$ .

**Longitude.** The trapezoid rule on  $n_\phi$  equally spaced points, which is exact for  $e^{i(m-m')\phi}$  only when  $|m - m'| < n_\phi$ . Analysing a band- $l_{\max}$  field needs  $|m|, |m'| \leq l_{\max}$ , hence

$$n_\phi \geq 2l_{\max} + 1. \quad (25)$$

At  $n_\phi = 2l_{\max}$  the orders  $m = +l_{\max}$  and  $m = -l_{\max}$  are the same discrete mode and cannot be separated. The library takes the smallest *fast* FFT length at or above the bound — restricted to  $2^a 3^b 5^c 7^d 11^e 13^f$  with  $e + f \leq 1$ , the set FFTW has codelets for — because the naive  $2l_{\max} + 2$  lands on  $514 = 2 \times 257$  at  $l_{\max} = 256$ , and 257 is prime. The smooth choice there is 520, at about 1.5% more FFT work, and the FFT is not the expensive stage (§9).

**Headroom.** A product of two band- $L$  fields has band  $2L$ , and  $\int |f|^2$  for band- $L$   $f$  integrates a degree- $2L$  quantity. Both want a grid resolving more than the fields do. `Grid::ForBand(L, nMax, oversampling)` expresses that distinction — the band of the fields against the resolution of the grid — rather than leaving a caller to compute a degree and hope; the  $3/2$  dealiasing rule is oversampling = 1.5 and 2 is exact for a single product.

## Part II

# Numerical methods

## 9 The transform

### 9.1 The algorithm

The forward transform of a field at upper index  $n$  is

$$f_{lm}^n = \sum_i w_i X_{lm}^n(\theta_i) F_m(\theta_i), \quad F_m(\theta) = \frac{2\pi}{n_\phi} \sum_j f(\theta, \phi_j) e^{-im\phi_j}, \quad (26)$$

so it is an FFT in longitude at each colatitude, followed by a weighted sum over colatitudes against the stored  $X$ . The inverse runs the same two stages in the other order. Both are separable exactly because (7) is a product of a function of  $\theta$  and  $e^{im\phi}$ .

**Cost.** The FFT stage is  $O(n_\theta n_\phi \log n_\phi)$  and the Legendre stage is  $O(n_\theta l_{\max}^2)$ , and at production sizes the second dominates completely. Measured at  $l_{\max} = 256$ ,  $n = 2$ : a forward transform takes 19.0 ms, of which the FFT stage is **0.43 ms** and the Legendre stage **18.6 ms** — a factor of 44. Any optimisation effort that does not address the Legendre stage is misdirected, and this number is the reason the library’s performance work concentrates entirely there.

**Where the Legendre stage’s time actually goes** is less obvious than it looks. Single-threaded it runs at about 19% of the read bandwidth one core can achieve, so it is not DRAM-bound; what binds it is load/store throughput. Per stored value the inner loop does one 8-byte read that reaches memory and then, from cache, a read of the FFT bin and a read-modify-write of the coefficient — four memory operations, only one of which is a miss. Three hand-optimised rewrites of the kernel (weight hoisted, real and imaginary parts split, raw pointers) moved it by less than a factor of two in either direction.

## 9.2 Real fields

A real-valued field exists only at  $n = 0$ , where  $f_{l,-m} = (-1)^m \overline{f_{lm}}$  by (15). The transform then uses a real-to-complex FFT and stores only  $m \geq 0$ , halving both the FFT work and the coefficient count. At  $n \neq 0$  the same reduction is *invalid*: it assumes the self-relation  $f_{l,-m}^N = (-1)^{m-N} \overline{f_{lm}^N}$ , which holds only when  $f$  is its own conjugate, and a spin-weighted field is its own conjugate only at  $N = 0$ . The library therefore forbids real-valued fields at nonzero upper index in the type system, and the transform throws if asked.

## 9.3 Transforming is a projection

A component at upper index  $N$  has no content below degree  $|N|$ , because no harmonic there exists. So the forward transform of an *arbitrary* field discards whatever that field had at low degree, and evaluating the result does not reproduce the input. It reproduces its projection onto the span of the harmonics that exist.

This is not an error and not a loss of accuracy: it is what the expansion means. Once a field is band-limited — for instance because it came from an inverse transform — the round trip is an identity to rounding. Example 06 shows both, and prints  $5 \times 10^{-1}$  for the first and  $4 \times 10^{-14}$  for the second.

# 10 Computing the $d$ -functions

This is where nearly all the arithmetic is, and where the library differs most from the reference texts.

## 10.1 Recursion in $l$ , not in $m$

D&T describe two schemes, both at fixed degree: their (C.121) recurs in  $m$  at fixed  $N$ , and (C.122) recurs in  $N$  at fixed  $m$ . For stability they iterate (C.121) upward from  $m = -l$  and downward from  $m = +l$  and meet at  $m = N \cos \theta$ , seeding each end with the closed forms (10)–(11).

The library recurs in *degree*, at fixed  $(N, m)$ . This is the scheme Woodhouse used, and the code is an evolution of his Fortran 77 implementation; D&T do not discuss it. The recursion is

$$d_{Nm}^l = f_1 d_{Nm}^{l-1} - f_2 d_{Nm}^{l-2}, \quad (27)$$

with

$$f_1 = \frac{(2l-1)[l(l-1)\cos\theta - Nm]}{(l-1)\sqrt{(l^2 - N^2)(l^2 - m^2)}}, \quad f_2 = \frac{l\sqrt{((l-1)^2 - N^2)((l-1)^2 - m^2)}}{(l-1)\sqrt{(l^2 - N^2)(l^2 - m^2)}}. \quad (28)$$

**Why this one.** The transform consumes the values one degree at a time, in ascending order, for a fixed  $(N, \theta)$  — see (26). A recursion in  $l$  produces them in exactly that order, so the same loop that fills the table can in principle feed the transform directly, and a truncated transform can stop early and pay for nothing it does not use. A recursion in  $m$  produces a whole degree at once and would have to be run to completion first.

**What it costs.** Of the coefficients in (28), only the  $\cos \theta$  term depends on the colatitude; the four square roots are  $\theta$ -independent and are read from a precomputed table of  $\sqrt{k}$  and  $1/\sqrt{k}$  — two vectors of  $2l_{\max} + 1$  entries, which is the whole of the auxiliary storage the recursion needs. The inner loop runs over  $m$  at fixed  $(l, N, \theta)$  with unit stride on three rows, which vectorises.

**The removable singularity.** At  $l = |N| + 1$  the factor  $1/(l - 1)$  in (28) is  $1/|N|$ , which is a division by zero at  $N = 0$ ; but  $N/(l - 1) = \text{sign}(N)$  there, and  $d^{l-2}$  does not exist, so that degree is taken by a separate one-term branch with the singular factor cancelled analytically. The same one-term form appears at the “critical degree”  $l = m_{\max} + 1$ , where the row stops growing.

## 10.2 The seed row and the boundary terms

The recursion needs starting values in two senses.

**The seed row** , at  $l = |N|$ , is a loop over  $m$  at a single degree. At  $l = |N|$  the closed form reduces to

$$P_{l,-m}^N|_{l=|N|} = \sqrt{\binom{2l}{l+m}} \left(\sin \frac{\theta}{2}\right)^{l-m} \left(\cos \frac{\theta}{2}\right)^{l+m}, \quad (29)$$

and the binomial row is generated exactly by the standard step  $\binom{2l}{k} = \binom{2l}{k-1}(2l - k + 1)/k$ . The row is at most  $2|N| + 1$  long — five entries for a rank-2 application — so this is not about cost; see §10.4.

**The boundary terms** are the values at  $m = \pm l$ , needed once per degree as the row grows. Evaluating (10) directly costs three  $\log \Gamma$  and an exponential each time, which at  $l_{\max} = 256$  is about  $1.3 \times 10^5$  transcendental evaluations per upper index per pass. They obey a one-term recursion in  $l$ : writing  $s = \sin(\theta/2)$ ,  $c = \cos(\theta/2)$ ,

$$\frac{P_{l,-l}^N}{P_{l-1,-(l-1)}^N} = \sqrt{\frac{2l(2l-1)}{(l-N)(l+N)}} s c, \quad (30)$$

valid for  $l > |N|$ , seeded at  $l = |N|$  where the square root is one and the value is  $s^{2|N|}$  for  $N \geq 0$  and  $c^{2|N|}$  for  $N < 0$ . Both boundaries ride on the same recursion, because the ratio depends on  $N$  only through  $N^2$  and because  $P_{ll}^N = (-1)^{N+l} P_{l,-l}^{-N}$ .

## 10.3 Why the recursion is the safer form

The closed form (10) is evaluated in logarithms, and it has to be:  $\sqrt{(2l)!/\cdots}$  is of order  $4^l$  and overflows long before the value it belongs to, which is bounded by one. The recursion (30) multiplies by a factor of order  $2sc = \sin \theta \leq 1$  at each step, so nothing large is ever formed and no  $\log \Gamma$  is needed. It is more robust than what it replaced, not less.

Measured against the closed forms over  $l_{\max} = 128$ ,  $|N| \leq 3$  and six colatitudes, the worst relative discrepancy is  $2 \times 10^{-13}$  in double and  $9 \times 10^{-17}$  in long double — very nearly the same multiple of  $\varepsilon$ , about 1000, in both, and growing linearly in  $l_{\max}$  rather than with the number of entries compared. That says the drift belongs to the *closed form*, which exponentiates a logarithm of size  $O(l \log 4)$  and loses bits in proportion. The recursion’s values are the better ones, and the test’s tolerance is written as  $20 l_{\max} \varepsilon$  for that reason.

## 10.4 The poles, and a data race

At  $\theta = 0$  and  $\theta = \pi$  the ratio (30) is  $0 \times \infty$ . No special case is needed: the seed is already the exact 0 or 1 that the closed form’s boundary branches return, and the factor is zero thereafter, so the recursion produces the right answer unaided.

The seed row (29) was moved off the closed form for a different reason. glibc’s `lgamma` writes the global `signgam`, and the table is filled from every thread, so the closed form is a data race there — benign, since nothing reads `signgam`, but the only genuine one the library contains. Generating the binomial row instead removes the last  $\log \Gamma$  from any threaded path, and the values are exact rather than merely accurate.

Two things follow that are worth recording. Grid construction is about 8% faster, which is what the arithmetic predicts: the boundary is two values per degree out of  $2l + 1$ , so however expensive a  $\log \Gamma$  is, it is being paid on under 1% of the table. And ThreadSanitizer cannot confirm the absence of the race, because GCC’s `libgomp` carries no TSan annotations and every OpenMP region is reported as racing with whatever ran before it; a Clang build against a `libomp` compiled with `LIBOMP_TSAN_SUPPORT` is the only way to get a real answer.

## 11 Storage and access

### 11.1 The table

Values are held as a flat array indexed by  $(N, i_\theta)$  then by  $(l, m)$ , contiguous in  $(l, m)$  for each  $(N, i_\theta)$  — so a degree’s row is a contiguous run, which is what the transform’s inner loop wants. Offsets are closed form (`GSHIndices::OffsetForDegree`), not looked up.

The table is large. At  $l_{\max} = 256$  with  $n_{\max} = 2$  and all upper indices it is  $5 \times 257 \times (257^2 - N^2)$  doubles, about 648 MB; at  $l_{\max} = 512$  it is 5.4 GB. This is why the grid is a value-semantic handle over shared immutable state: copying one is a pointer copy, and `sizeof(Grid)` is 16 bytes. Before that change, twenty copies of a  $l_{\max} = 128$  grid cost 1638 MB; afterwards, nothing.

### 11.2 The row supplier

Both transform loops take the row for a degree from an accessor rather than walking a single iterator across the whole block. The only thing either loop needs is *a contiguous run of values in  $(l, m)$  order for one  $(N, i_\theta)$* , and asking for it one degree at a time costs a few integer operations against a loop of length  $(2l + 1)k$ .

That seam exists so the values can come from somewhere other than a table. A second implementation runs the recursion into per-thread scratch inside the transform, and a grid can be asked for it at construction. It builds no table at all: construction at  $l_{\max} = 256$  goes from 0.248 s and 648 MB to 0.015 s and nothing.

**It is slower, and the measurement is worth keeping.** On an 8-core laptop the generated path is  $1.4\text{--}3.2\times$  slower than the stored one in every configuration measured. Batching narrows the gap, since generation amortises over a chunk exactly as a table stream does, but *threads do not*: the ratio at eight threads is the ratio at one. The premise that the stored path is pinned at the DRAM roof while a generated one scales with cores is not visible on eight cores, and the deployment target has more memory bandwidth per core, not less. What survives is the memory saving, which is large, and a NUMA argument that only a two-socket machine can settle.

## 12 Batching, chunking and threading

### 12.1 The batch

The transform’s primitive is “transform  $k$  fields sharing a grid, a degree and an upper index”, with the single field as  $k = 1$ . What is being amortised is the Wigner block for that upper index, which is why fields at different upper indices cannot batch together — for a rank-2 tensor that means batching over radii within each  $N$ , not across components.

A batch is described by (count, stride, dist) rather than by contiguity: element  $j$  of field  $k$  lives at  $j \cdot \text{stride} + k \cdot \text{dist}$ . This is FFTW’s advanced-interface form, and it costs nothing at the stage that actually reads the caller’s layout, so it covers both arrangements the field layer produces:

|                                          | stride                  | dist      |
|------------------------------------------|-------------------------|-----------|
| components stored component by component | 1                       | FieldSize |
| components stored point by point         | $n_{\text{components}}$ | 1         |

**The descriptor’s uniformity is a constraint on tensor storage** , which was not obvious in advance. Its members must be evenly spaced, and in flat multi-index order the stored components sharing one upper index are scattered at no fixed spacing as soon as there is any symmetry. So a tensor’s buffer is ordered *by upper index*, which makes each group a contiguous run on both the field side and the coefficient side. Ordered any other way, the batching the primitive exists for would be unreachable from the layer it was built for.

**What batching is worth.** About  $2.3\times$  at the optimum, not  $k$ . The gain saturates because the stage is bound by per-element load/store work rather than by table bandwidth, and batching removes  $(k - 1)/k$  of the *table* reads while leaving that untouched. It also *reverses* beyond an optimum: the coefficient array is  $k$  times larger, and once it stops fitting in cache it is streamed from DRAM too.

### 12.2 Chunking

Because of that reversal, a batched call processes a large batch in chunks of an internally chosen size rather than handing the caller’s  $k$  straight to the inner loop. The chunk is

$$\text{chunk} \approx \frac{\text{cache}}{2 \times \text{copies} \times \text{bytes per coefficient block}}, \quad (31)$$

rounded to nearest, the factor of two being headroom for the streamed Wigner row and the FFT output.

**“Copies” is not the thread count in both directions** , and getting that wrong cost a factor of two. The forward transform gives every thread a private accumulator of  $\text{chunk} \times \text{coefficientSize}$ , so its copies are its threads; the inverse gathers one block, shared and read-only, so it has one copy however many threads read it. Serving both with the thread count starved the inverse — at  $l_{\text{max}} = 256$ ,  $k = 8$  on eight threads it returned a chunk of one where the whole batch fits, and the correction measured  $2.0\times$  on that direction.

One thing the rule still cannot see: on a multi-CCD or multi-socket machine a shared block is pulled into each cache domain that touches it, so the inverse’s single copy is really one per domain.

### 12.3 Threading

Threading is an explicit per-call policy and defaults to sequential. The library creates threads because it was asked to and not because it can, and the rule it keeps is that **exactly one level threads**: an operation asked to run in parallel from inside an existing parallel region runs sequentially instead, so a caller parallelising over slices, components or realisations cannot nest with the transform, and neither can the table build.

The decomposition is over colatitudes, and the two directions differ. The inverse transform’s colatitudes write disjoint rows and share only read-only input, so they divide with no reduction at all. The forward transform’s colatitudes all contribute to every coefficient, so each thread accumulates into a private buffer and the partial sums are added at the end — partitioned rather than serialised: each thread sums every thread’s partials for its own block of coefficients.

**Measured** , at  $n = 2$ , complex, per call:

|                            | 1 thread | 2            | 4            | 8                              | 16                             |
|----------------------------|----------|--------------|--------------|--------------------------------|--------------------------------|
| $l_{\max} = 128$ , forward | 1.75 ms  | $1.89\times$ | $2.98\times$ | <b><math>4.02\times</math></b> | $3.13\times$                   |
| $l_{\max} = 128$ , inverse | 1.77 ms  | $1.83\times$ | $3.07\times$ | $4.15\times$                   | <b><math>4.57\times</math></b> |
| $l_{\max} = 256$ , forward | 12.16 ms | $1.63\times$ | $2.51\times$ | <b><math>2.68\times</math></b> | $2.23\times$                   |
| $l_{\max} = 256$ , inverse | 12.54 ms | $1.69\times$ | $2.64\times$ | <b><math>2.98\times</math></b> | $2.97\times$                   |

Sixteen threads is not better than eight and is often worse: the machine has eight cores and sixteen hardware threads, and for memory-bound work the second thread on a core adds contention rather than throughput. Callers should ask for cores.

At  $l_{\max} = 128$  on eight threads the stage reaches 39–40 GB/s against a measured machine roof near 42, so it is genuinely bandwidth-bound there and nothing further can be won without reducing the traffic itself.

## 13 What the type system enforces

The field algebra is built so that a statement which is not mathematically meaningful does not compile. What follows is the list, with the reason in each case.

### 13.1 The index algebra

Every node — an owning field, a view, or an expression — carries its upper index as a compile-time constant, and the operators enforce §2:

| expression                                       | upper index | requires            |
|--------------------------------------------------|-------------|---------------------|
| $f + g, f - g$                                   | $N$         | equal upper indices |
| $fg$                                             | $N_f + N_g$ | —                   |
| $f/g$                                            | $N_f$       | $N_g = 0$           |
| $\overline{f}$                                   | $-N_f$      | —                   |
| $ f ,  f ^2$                                     | 0           | —                   |
| $\operatorname{Re} f, \operatorname{Im} f, F(f)$ | 0           | $N_f = 0$           |
| $\int f \, d\Omega$                              | —           | $N_f = 0$           |

Three of these were wrong in the layer this replaced, and they are worth naming. **Conjugation reverses the upper index**: a quantity carrying  $e^{-iN\psi}$  has a conjugate carrying  $e^{+iN\psi}$ . **Real and imaginary parts exist only at  $N = 0$** , because for  $N \neq 0$  the split is not preserved by a frame rotation and neither part is a component of anything. **Nonlinear functions likewise**, and integration with them, since  $\int f \, d\Omega$  vanishes identically unless  $N = 0$ .

The constraints are **requires**-clauses on the free operators, so an unlawful combination is an overload-resolution failure at the call site rather than an error inside a node — which also means the negative cases can be tested.

### 13.2 Real-valuedness

A node may be real-valued *only* at  $N = 0$ . Real-valuedness is not preserved by the frame rotation, so it is not a property any component of any tensor can have elsewhere; a library able to represent one could represent something that does not exist. The constraint is closed under every operation in the algebra —  $\text{Re}$ ,  $\text{Im}$ ,  $|f|$ ,  $|f|^2$  and  $F(f)$  all land at  $N = 0$  precisely because that is the only place their results could be real — so it is checked once, on the concept, and never re-derived.

### 13.3 Laziness, and the aliasing theorem

Operators build expression nodes; nothing is evaluated until a value is read. Every node in the layer is **pointwise and index-preserving**: the value at  $(i_\theta, i_\phi)$  reads its operands only at  $(i_\theta, i_\phi)$ . Hence

$u = \text{expr}$  is safe when  $u$  occurs in  $\text{expr}$ , and needs no temporary.

That is a theorem about the node set rather than a property of any one operator, which is why the invariant is stated in the stronger “index-preserving” form: a re-indexing view — a longitude shift, a transpose of the grid — would be pointwise in the loose sense and would break it. Operations that are genuinely not pointwise (gradients, raising and lowering, radial derivatives) live in the spectral layer for this reason and not by accident.

Operand storage follows the value category: a terminal passed as an lvalue is held by reference, everything else by value. The residual hazard — an lvalue terminal destroyed while an expression referring to it is alive — is the same one Eigen has and is not removable in C++.

### 13.4 Which components a tensor stores

Both reductions of §5 are computed by one **constexpr** walk of the group generated by the permutation symmetry together with, for a real tensor, the negation. Reaching a component twice by different routes is the interesting case rather than a failure: equating the two routes produces a constraint.

| the two routes differ by    | the orbit is     |
|-----------------------------|------------------|
| a sign, same conjugation    | identically zero |
| conjugation, signs equal    | real             |
| conjugation, signs opposite | purely imaginary |

The first arises without reality at all — it is the vanishing diagonal of an antisymmetric tensor. The others are the self-paired components, which always sit at  $N = 0$ , and are therefore exactly where a real-valued field is admissible.

One consequence reaches the interface: a component accessor cannot return one type. A stored component is a view, a symmetric-permutation relative is the same view, an antisymmetric relative is that view negated, a reality relative is its conjugate at the reversed upper index, and a vanishing orbit has nothing to return. Traversal over all  $3^p$  components is therefore a compile-time loop, not a runtime one.

### 13.5 The two domains are not symmetric

On the spatial side a derived component is pointwise —  $T^{-\alpha} = (-1)^N \overline{T^\alpha}$  — so a view plus an expression node gives it. On the spectral side the corresponding relation is (15),  $T_{l,-m}^{-N} = (-1)^m \overline{T_{lm}^N}$ , which *reverses the order index*. That is not a view over anything, so a tensor expansion exposes only its stored components and deriving is done in the spatial domain. This is the one place where the mirror between field and expansion breaks.

## 14 Validation

What each oracle actually pins, since a test suite that all passes is uninformative about which facts are load-bearing.

| oracle                             | what it pins                                                                                                                                                                                                                            |
|------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>CheckWignerConvention</code> | That the stored value is $X_{lm}^N$ with the upper index first, against all nine $l = 1$ values of D&T (C.115). Discriminating: the four entries with $N - m$ odd distinguish $d_{Nm}^l$ from $d_{mN}^l$ .                              |
| <code>CheckLegendre</code>         | The $N = 0$ row against <code>std::sph_legendre</code> , which fixes the orthonormal scaling and the Condon–Shortley phase.                                                                                                             |
| <code>CheckAdditionTheorem</code>  | (12) over all $ N ,  N'  \leq 40$ at $l_{\max} = 40$ , to $1000\varepsilon$ . The strongest single check on the recursion: boundary values feed the two-term recursion at every higher degree, so an error anywhere propagates into it. |
| <code>CheckWignerBoundary</code>   | The boundary recursion and the binomial seed row against the closed forms they replaced, to $20l_{\max}\varepsilon$ — a tolerance argued from where the drift actually is (§10).                                                        |
| <code>CheckCoeff2Coeff</code>      | Coefficient-to-coefficient round trips.                                                                                                                                                                                                 |
| <code>TestRealFieldSymmetry</code> | (14) at $n = 2$ through complex transforms, and the rejection of real transforms at nonzero upper index.                                                                                                                                |
| batched transform tests            | A batch of $k$ against $k$ separate calls, at <i>exact</i> equality. Batching widens the inner loop without reordering any sum, so any difference at all would mean it had been restructured rather than widened.                       |
| generated-path tests               | The generating supplier against the stored table, again at exact equality: same recursion, same order, same rounding, only the storage differs.                                                                                         |
| reality tests                      | (13) at every component of every tensor, and that the stored set costs $3^p$ reals per point.                                                                                                                                           |
| $\tilde{\mathcal{O}}$ tests        | (17), both of them.                                                                                                                                                                                                                     |

**On measuring this library.** The development machine’s clock scales under load, and the noise floor between back-to-back runs of identical code is about 10%. Comparisons must interleave the two versions in one process; figures from different sessions are worth less still. Within-run ratios — the stage split, the batching curve — are unaffected, because their terms are measured moments apart.

## 15 What has not been done, and why

Recorded so that the absences read as decisions.



**The symmetry reductions of §4** would cut the table fourfold. They are not implemented because in the regime that matters they would buy little: batched, with a chunk that fits, the Legendre stage runs at about a fifth of the memory roof and under a tenth of compute peak, so halving the traffic addresses neither binding constraint. Unbatched they would be worth roughly  $2\times$ ; batching already took that.

**Reduced-precision storage** is in the same position, and for the same reason. It also needs a tighter round-trip oracle than exists before the experiment would mean anything.

**A transform-major layout with a GEMM** is the one restructure aimed at what actually binds — the per-element load/store count — and it subsumes the question of how the forward transform should decompose above about sixteen threads, since a per- $m$  decomposition has disjoint outputs and needs neither accumulator nor reduction. It is the natural next piece of performance work.

**Polar truncation**, skipping the  $(m, \theta)$  pairs where  $d_{Nm}^l$  is negligible — it falls off like  $\sin^{|m|} \theta$  — is the one option that makes the problem smaller rather than making the machine work harder, and it reduces storage and arithmetic together. It is not implemented and probably dominates both of the reductions above.

**The two open conventions** of §1 want an answer from the literature rather than from a test.